OGC API JSON-FG

Geonovum Standards workshop

Auteur Pieter Bresters / Joost Farla

Datum Wednesday 25 June 2026

Program

10.00 - 10.15	Welcome and introductions
10.15 - 10.30	Introduction to OGC in general
10.30 - 11.00	OGC JSON-FG in detail
11:00 - 11.15	Break
11.15 - 11.30	Validating a JSON-FG file
11.30 – 12.00	Hands-on session: Get started with JSON-FG yourself
12.00 - 12.45	Lunch break
12.45 - 13.30	Brainstorm pilot project with data exchange
13.30 - 14.45	Hands-on session: Get on with JSON-FG yourself
14.45 - 15.00	Wrap-up and closing
15.00 - 16.00	Optional extension: You are welcome to stay longer for further support from our API experts.

JSON FG is an OGC-standard

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About The Open Geospatial Consortium (OGC):

- International standards organization for geospatial data and services
- Develops open, consensus-based standards to make geospatial information findable, accessible, interoperable and reusable
- Members include government agencies, companies, research institutes and universities worldwide
- Maintains widely used standards such as WMS, WFS, GeoPackage and the modern OGC API family

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OGC Releases the Features and Geometries JSON (JSON-FG) Standard

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Announcement • May 12, 2026

OGC Publishes openEO as a New Community Standard

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Why is Geonovum supporting JSON-FG

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- Request from Dutch Kadaster / Sweco
- Geonovum was involved in developing JSON-FG
- It contributes to a higher goal: encouraging the use of OGC API's, because API's benefit from this new standard.
- It will be used for new BGT data exchange.

What is JSON-FG

- Features and Geometries JSON
- It is an encoding for special geometries
- Extension to GeoJSON
- Backwards compatible

- Established on April 30, 2026:
- https://www.ogc.org/standards/json-fg/

https://docs.ogc.org/is/21-045r1/21-045r1.html

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Submission Date: 2025-06-02

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Publication Date: 2026-04-30

External identifier of this OGC® document: http://www.opengis.net/doc/IS/json-fg-1/1.0

Internal reference number of this OGC® document: 21-045r1

Version: 1.0.0

Category: OGC® Implementation Standard

Editor: Clemens Portele, Panagiotis (Peter) A. Vretanos

OGC Features and Geometries JSON - Part 1: Core

Why JSON-FG

- GeoJSON is widely used (OGC API's)

- But, it has its limitations:
  - only WGS84
  - no 3D
  - no standard for datetime attributes
  - no curves, arcs nor measures
  - no reference to schemas and feature type

JSON-FG has it!

Mandatory Extensions

- Ability to use Coordinate Reference Systems (**CRS**) other than WGS 84;
- Ability to encode **temporal** characteristics of a feature;
- A statement identifying the **JSON-FG conformance** classes supported by a JSON-FG feature collection, feature, or geometry.

JSON-FG has it!

Optional Extensions with support for:

- **3D**: solids and prisms as geometry types;
- **Complex geometry**:
  - circular arcs
  - compound curves
  - curve polygons;
- Support for **measure** values;
- The ability to declare the **type** and **schema** of a feature.

CRS

WGS84 (longitude, latitude)

RD (x, y)

Use **coordRefSys**-element:

- RD "coordRefSys": "http://www.opengis.net/def/crs/EPSG/0/28992",
- WGS84 "coordRefSys": "http://www.opengis.net/def/crs/OGC/0/CRS84",

Place or Geometry

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- Use place element when special FG-data types are used or when CRS is not WGS84
- Use geometry element when CRS is WGS84 and no special FG-data types are used
- When place and geometry are both used, they should not be identical

Temporal data

- Use **time**-element

- Instants: "time": {"instant": "2022-10-21T18:00:00Z"}

  - Date: “2002-12-31”
  - Timestamp: “2002-12-31T23:00:00Z”

- Intervals: "time": { "interval": ["2002-12-31T23:00:00Z", "2003-12-31T23:00:00Z"] } }

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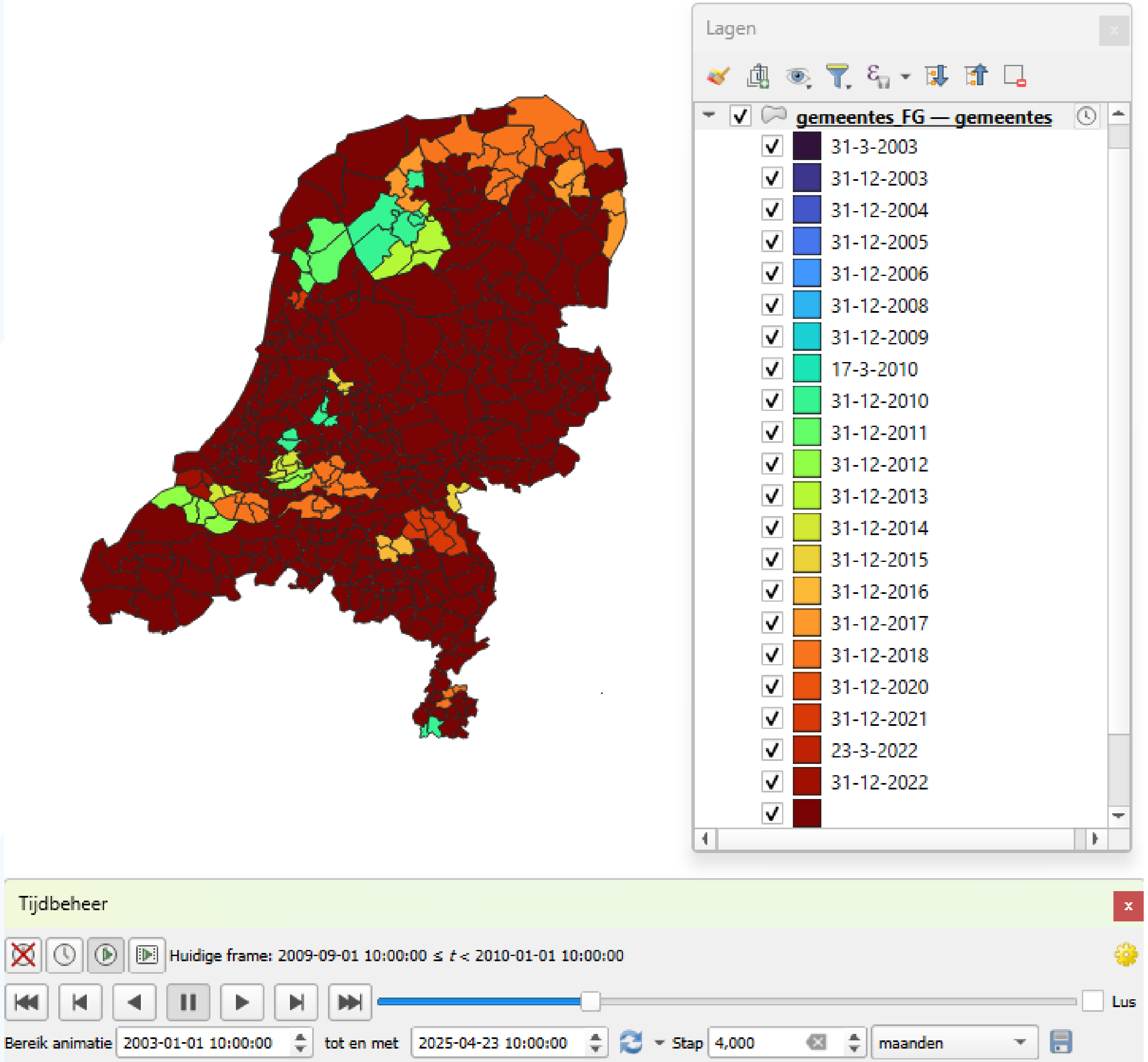
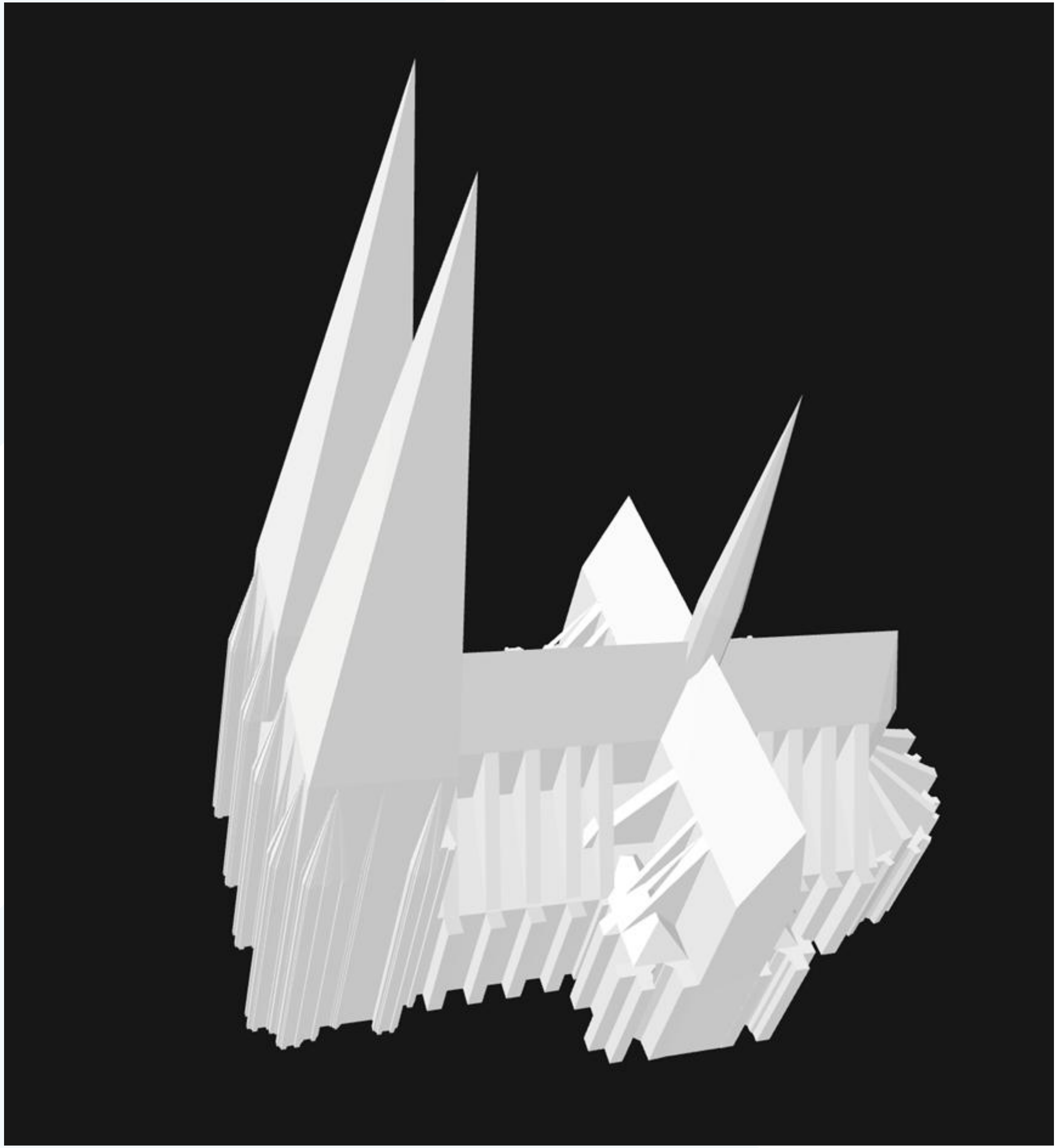
Temporal data



3D

- Polyhydron
- Multi-Polyhydron



3D

- Prism (polygon or circle with upper and lower limit) 2,5 D
- Muti-prism

Complex geometry

- **CircularString**
  - a sequence of an odd number of points with a minimum of three points
- **CompoundCurve**
  - a sequence of circular strings and/or line strings
- **CurvePolygon**
  - a sequence of closed rings, build up by line strings, circular strings or line strings
  - The first ring is the exterior boundary and, all other rings are interior boundaries
- **MultiCurve**
  - a set of line strings, circular strings or compound curves
- **MultiSurface**
  - a set of polygons or curve polygons

Complex geometry: CircularString

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Arc


```json
{
    "conformsTo": [
        "http://www.opengis.net/spec/json-fg-1/1.0/conf/core"          ,
        "http://www.opengis.net/spec/json-fg-1/1.0/conf/circular-arcs"
    ],
    "coordRefSys": "http://www.opengis.net/def/crs/OGC/0/CRS84",
    "type": "CircularString",
    "coordinates": [
        [4.5368482, 51.5913298],
        [4.5371346, 51.591267],
        [4.5373782, 51.591154]
    ]
}
```


Circle

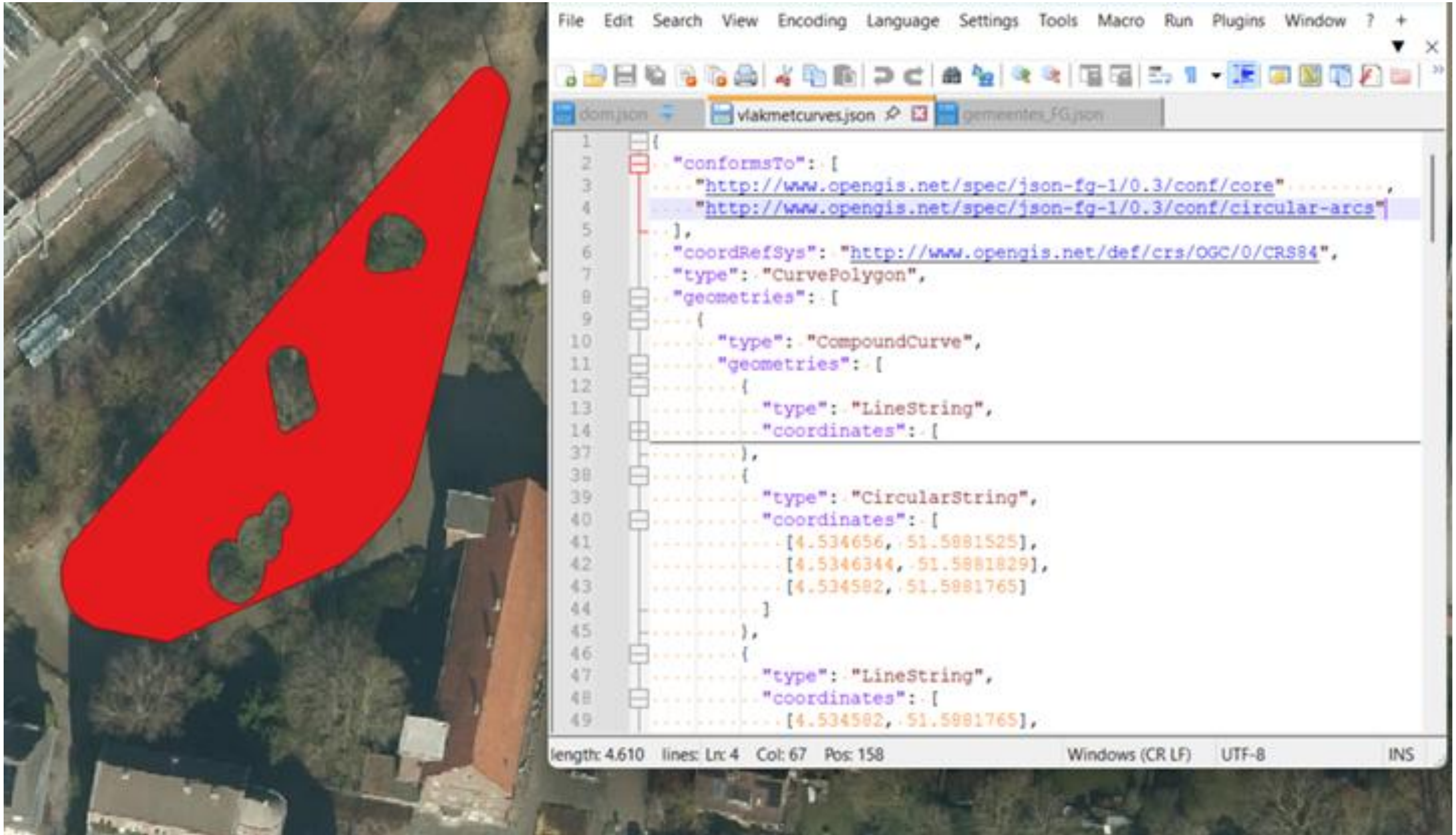

```json
{
    "conformsTo": [
        "http://www.opengis.net/spec/json-fg-1/1.0/conf/core"          ,
        "http://www.opengis.net/spec/json-fg-1/1.0/conf/circular-arcs"
    ],
    "coordRefSys": "http://www.opengis.net/def/crs/OGC/0/CRS84",
    "type": "CircularString",
    "coordinates": [
        [4.5390810, 51.5902305],
        [4.5390200, 51.5902275],
        [4.5390248, 51.5901896],
        [4.5390857, 51.5901925],
        [4.5390810, 51.5902305]
    ]
}
```

Complex geometry:

CompoundCurve

```json
{
  "conformsTo": [
    "http://www.opengis.net/spec/json-fg-1/1.0/conf/core"          ,
    "http://www.opengis.net/spec/json-fg-1/1.0/conf/circular-arcs"
  ],
  "coordRefSys": "http://www.opengis.net/def/crs/OGC/0/CRS84",
  "type": "CompoundCurve",
  "geometries": [
    {
      "type": "LineString",
      "coordinates": [
        [4.5362932, 51.5906989],
        [4.5362737, 51.5906938]
      ]
    },
    {
      "type": "CircularString",
      "coordinates": [
        [4.5362737, 51.5906938],
        [4.5362567, 51.5906829],
        [4.5362591, 51.5906678]
      ]
    },
    {
      "type": "LineString",
      "coordinates": [
        [4.5362591, 51.5906678],
        [4.5363324, 51.5906866]
      ]
    },
    {
      "type": "CircularString",
      "coordinates": [
        [4.5363324, 51.5906866],
        [4.5363173, 51.5906983],
        [4.5363037, 51.5906992]
      ]
    }
  ]
}
```

Complex geometry: CurvePolygon

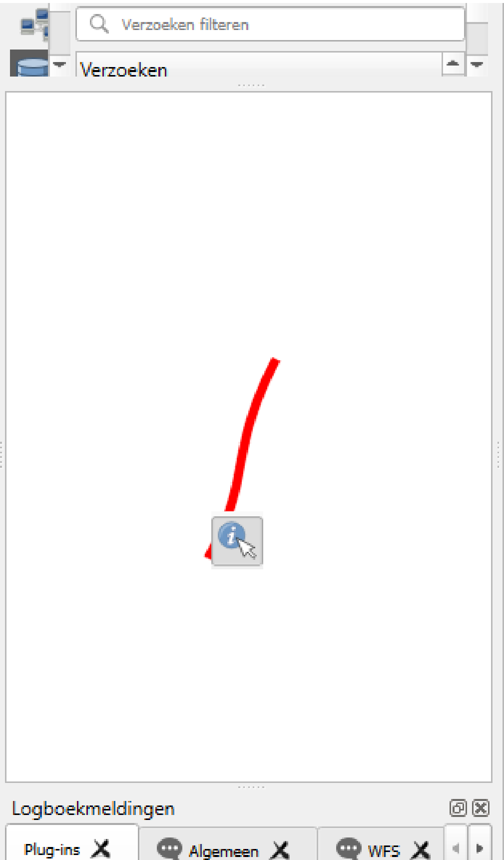


Measure

- Linear referencing:

Measure


```json
{
  "type": "Feature",
  "id": "371304103613255A",
  "conformsTo": [
    "http://www.opengis.net/spec/json-fg-1/1.0/conf/core"    ,
    "http://www.opengis.net/spec/json-fg-1/1.0/conf/measures"
  ],
  "coordRefSys": "http://www.opengis.net/def/crs/OGC/0/CRS84",
  "measures": {
    "enabled": true,
    "unit": "km",
    "description": "Kilometers along the road based on the kilometer markers. Positions between two physical markers are interpolated."
  },
  "geometry": null,
  "place": {
    "type": "LineString",
    "coordinates": [
      [7.9379077, 52.2841795, 227.396],
      [7.9386618, 52.2856218, 227.227],
      [7.9389695, 52.2862999, 227.149],
      [7.9395120, 52.2876169, 226.998],
      [7.9397562, 52.2883001, 226.920],
      [7.9401802, 52.2896330, 226.769],
      [7.9405226, 52.2910055, 226.614],
      [7.9412150, 52.2945101, 226.222],
      [7.9415283, 52.2959768, 226.057],
      [7.9416385, 52.2964391, 226.005],
      [7.9421299, 52.2981305, 225.814],
      [7.9426214, 52.2995303, 225.655],
      [7.9429592, 52.3003909, 225.556],
      [7.9434566, 52.3015636, 225.422],
      [7.9438507, 52.3024014, 225.325],
      [7.9443588, 52.3034069, 225.208]
    ]
  },
  "time": {"instant": "2022-10-21T18:00:00Z"},
  "properties": {
    "road.title": "A1",
    "road.class": "A",
    "road.number": 1,
    "num": "8",
    "type": "Abschnitt",
    "lr_start": 0,
    "length_m": 2188,
    "adminarea.name": "Bundesrepublik Deutschland",
    "adminarea.knz": "00000000000",
    "netzstand": "2022-10-21T18:00:00Z"
  }
```

Object	Waarde
road-segment	
adminarea.name	Bundesrepublik Deutschland
(Afgeleid)	
(geklikt op coördinaat X)	7,9395°
(geklikt op coördinaat Y)	52,2863°
Dichtstbijzijnde X	7,939°
Dichtstbijzijnde Y	52,2865°
Dichtstbijzijnde punt M	227,149
Dichtstbijzijnde punt X	7,939°
Dichtstbijzijnde punt Y	52,2863°
Geïnterpoleerde M	227,1292295
Hoekpunten	16
Lengte (Cartesiaans)	0,0204 deg
Lengte (Ellipsoïde — E...	2188,8242 m
Nummer dichtstbijzijn...	3
Object-ID	0
eersteX	7,9379°
eersteY	52,2842°
laatsteX	7,9444°
laatsteY	52,3034°
(Acties)	
id	371304103613255A
road.title	A1
road.class	A
road.number	1
num	8
type	Abschnitt
lr_start	0
length_m	2188
adminarea.name	Bundesrepublik Deutschland
adminarea.knz	00000000000
netzstand	21-10-2022 18:00:00 (UTC)

Schema

- featureType for whole json-file or per feature

```
{
"type": "FeatureCollection",
"conformsTo": ["http://www.opengis.net/spec/json-fg-1/0.3/conf/core", "http://www.opengis.net/spec/json-fg-1/0.3/conf/types-schemas"],
"featureType" : "gemeentes",
"coordRefSys": "http://www.opengis.net/def/crs/EPSG/0/28992",
"features" : [
{ "type": "Feature", "properties": { "code": "0258", "naam": "Kesteren", "begingeldigheid": "2002-12-31", "eindgeldheid": "2003-03-31" },
{ "type": "Feature", "properties": { "code": "0178", "naam": "Rijssen", "begingeldigheid": "2002-12-31", "eindgeldheid": "2003-12-31" },
```

```
{
    "type": "FeatureCollection",
    "coordRefSys": "http://www.opengis.net/def/crs/EPSG/0/5555",
    "conformsTo" : [ "http://www.opengis.net/spec/json-fg-1/0.3/conf/core", "http://www.opengis.net/spec/json-fg-1/0.3/conf/types-schemas'
    "features": [
        {
            "type": "Feature",
            "featureType": "Building",
            "id": 1672388,
            "geometry": null,
            "properties": {
        },
        {
            "type": "Feature",
            "featureType": "BuildingPart",
            "id": 2300938,
            "geometry": null,
            "properties": {
            "place": {
                "type": "Polyhedron",
                "coordinates": [
                    [
                        [
                            [
                                [ 356488.442, 5645311.721,  97.818 ],
                                [ 356487.962, 5645311.720,  97.818 ],
```

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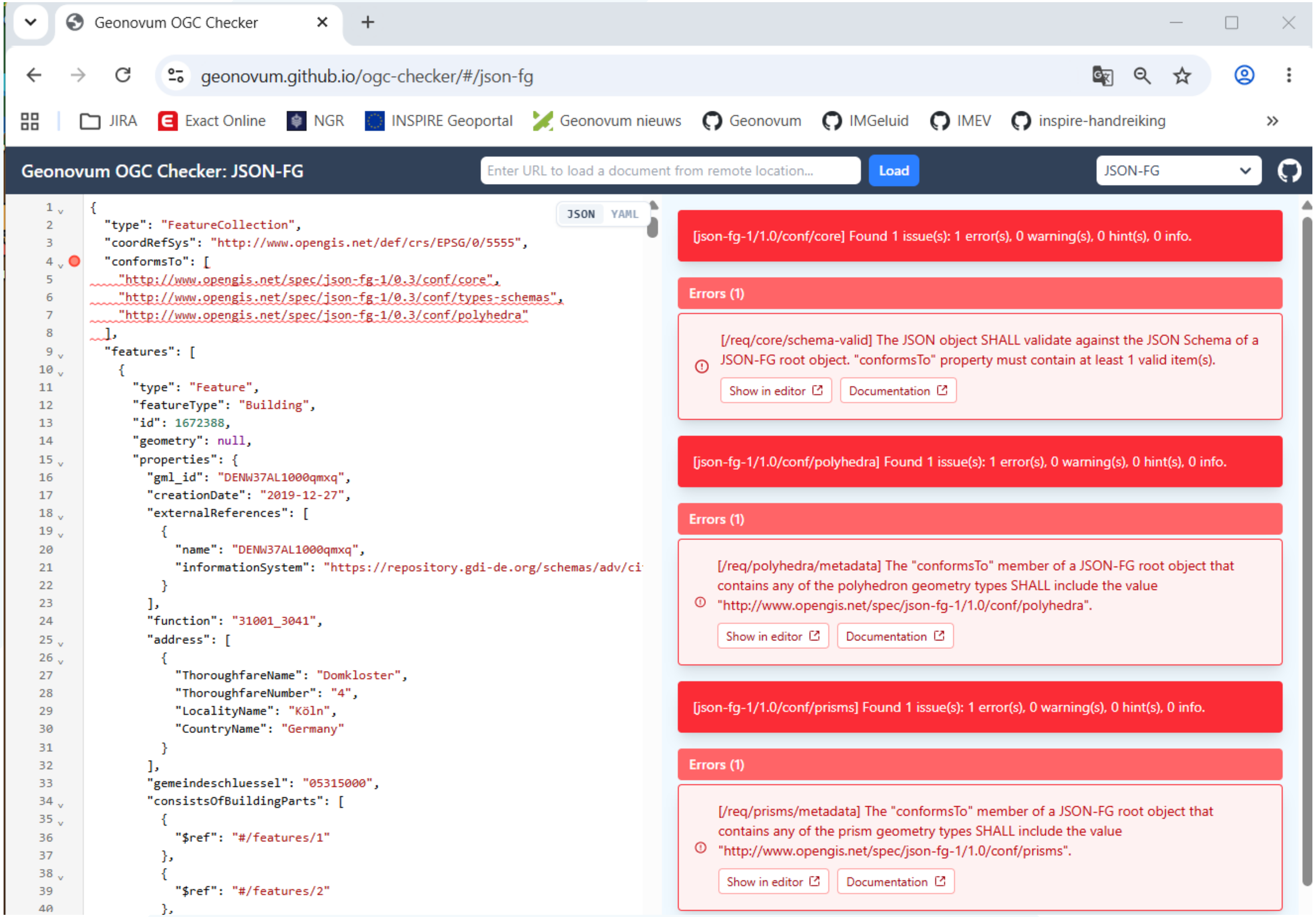
Profiles

- HTTP: in HTTP header "Link" using the link relation "profile". (value of link = URI of profile)
- Or in JSON-FG document

- *RFC 7946 (GEOJSON)*
  Link: <http://www.opengis.net/def/profile/OGC/0/rfc7946>; rel="profile"
- *JSON-FG*
  Link: <http://www.opengis.net/def/profile/OGC/0/jsonfg>; rel="profile"
- *JSON-FG with improved support for GeoJSON readers*
  Link: <http://www.opengis.net/def/profile/OGC/0/jsonfg-plus>; rel="profile"

Validations



Get started with JSON-FG yourself

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- Exercises are available on Github
  - https://github.com/Geonovum/ogc-api-workshops/
- View a JSON-FG file in browser or JSON-editor
  - CRS
  - 3D (https://dev.3dgi.xyz/jsonfg-viewer/#)
  - Curves
  - Measures
- Validate JSON-FG
- Lunch break
- Brainstorm pilot project with data exchange
- Load JSON-FG into QGIS
- Create a time-slider on a JSON-FG in QGIS
- Create a 3D-scene in QGIS
- Inspect a road segment with measures in QGIS

Lessons learned during the workshop

1. An overview of the JSON-FG standard
2. Dealing with time and other CRS's
3. The extra data types:
    1. 3D
    2. Curves
    3. Measures
4. Validation
5. Loading JSON-FG into GIS software.

Follow up

- Presentations and handson remain available on Github.
- Ask your questions via the GeoForum.
- Feedback from participants?
- More workshops to follow, see Geonovum agenda.
- More information:
  - https://docs.ogc.org/is/21-045r1/21-045r1.html
  - https://github.com/opengeospatial/ogc-feat-geo-json

Thank you!

Geonovum

T 033 460 41 00

E info@geonovum.nl

I www.geonovum.nl

bezoekadres

Barchman Wuytierslaan 10

3818 LH Amersfoort

postadres

Postbus 508

3800 AM Amersfoort